

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

593000506 - Distributed Systems For Iot

DEGREE PROGRAMME

59AH - Master Universitario En Internet Of Things (IOT)

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	593000506 - Distributed Systems For Iot
No of credits	4.5 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	59AH - Master Universitario en Internet Of Things (Iot)
Centre	61 - E.T.S De Ing. De Sistemas Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Silvia Alba Uribe Mayoral	4218	silviaalba.uribe@upm.es	Sin horario. Please, send and email to schedule an appointment.
Gustavo Adolfo Hernandez Peñaloza (Subject coordinator)	4408	gustavo.hernandez.penalaza@upm.es	Sin horario. Please, send and email to schedule an appointment.

Juan Antonio Rodrigo Ferran	4201	juanantonio.rodrido@upm.es	Sin horario. Please, send and email to schedule an appointment.
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* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Programming languages, computer networks, operating systems

4. Skills and learning outcomes *

4.1. Skills to be learned

CB08 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CE.05 - Diseñar y desarrollar sistemas distribuidos para dar soporte a aplicaciones IoT, evaluando las tecnologías mas apropiadas de acuerdo con los diferentes contextos de aplicación como son dispositivos móviles, sistemas en tiempo real o sistemas ubícuos

CE.06 - Analizar el rendimiento, disponibilidad, escalabilidad y fiabilidad de los sistemas distribuidos empleados en aplicaciones IoT

CE.08 - Diseñar y desarrollar soluciones tecnológicas para implementar servicios IoT capaces de interactuar con diferentes fuentes de información y dispositivos distribuidos incluyendo el diseño de estructuras de intercambio de

información eficientes

CG01 - Los alumnos demostrarán tener una visión del estado actual, las necesidades y los problemas que se plantean en el mundo de la IoT, así como de las arquitecturas y estándares más utilizados

CG02 - Los alumnos serán capaces de aplicar métodos y tecnologías avanzadas que les permitan abordar necesidades y problemas en aplicaciones IoT

CT.01 - Capacidad de uso de la lengua inglesa para el trabajo en contextos internacionales

4.2. Learning outcomes

RA12 - Coordinate IoT microservices using replicated state-machines services like Zookeeper.

RA10 - Design and implement a IoT microservice architecture based on Kafka stream processing platform.

RA40 - To identify new application domains for IoT.

RA11 - Choose the best replica consistency type for a IoT microservice.

RA13 - Apply CAP theorem principles to choose between availability and consistency

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Software that coordinates a set of connected computers in a communication network to get a certain goal is denominated Distributed System. This course will study different models of interactions among devices, computing services and data services used in IoT, Cloud Computing, Blockchain and Big Data. It will also study the scalability and availability techniques to replicate services with different degrees of consistency. First, typical Distributed System models and architectures are presented. Then, indirect communication publish/subscribe paradigm among processes of a distributed system will be explained, in order to introduce Stream Processing as a real time data processing. Next, Blockchain Ethereum architecture foundations will be shown for a non fault tolerant scenario (one server). The replica consistency and CAP theorem will be explained in order to introduce both Raft and the Fault Tolerant design of Blockchain-Ethereum. In the lab, we will develop distributed applications made up of hardware and software elements real and simulated with NodeRed. Then, the MQTT protocol will be described to use it along with NodeRed. Finally, we will move on to the Kafka Stream Processing tool, and we end with a short example of

Ethereum Blockchain. Finally, a fully distributed system for Artificial intelligence models training.

5.2. Syllabus

1. Introduction

- 1.1. Definition
- 1.2. Abstractions
- 1.3. Examples
- 1.4. Models

2. Indirect Communication. Publisher-subscriber

- 2.1. Definition
- 2.2. Examples
- 2.3. Properties
- 2.4. Programming Model
- 2.5. Implementation

3. Stream processing

- 3.1. Case study definition
- 3.2. Streaming aggregation
- 3.3. Event sourcing
- 3.4. Separating DB Reads and Writes
- 3.5. Immutable Facts and the Source of Truth
- 3.6. Using Append-Only Streams of Immutable Events

4. Blockchain. Ethereum. Non Fault Tolerant Approach (1)

- 4.1. Introduction
- 4.2. Cryptography
- 4.3. Clients. Transactions
- 4.4. Servers. Miners. Transaction validation. Block execution
- 4.5. Smart contracts

- 4.6. World state
- 5. Replica consistency and CAP theorem
 - 5.1. Model
 - 5.2. Atomic consistency
 - 5.3. Sequential consistency
 - 5.4. Causal Consistency
 - 5.5. Eventual consistency
 - 5.6. CAP theorem
- 6. Raft. Atomic consistency
 - 6.1. Introduction
 - 6.2. Model
 - 6.3. Raft algorithm basics
 - 6.4. Leader election
 - 6.5. Log replication
 - 6.6. Safety
- 7. 7 Blockchain. Ethereum. Fault Tolerant Approach (2)
 - 7.1. Peer-to-peer network
 - 7.2. Proof of work. Eventual consensus
 - 7.3. Eventual consistency replication
- 8. Federated Systems
 - 8.1. Introduction
 - 8.2. Architectures
 - 8.3. Implementations and tools
 - 8.4. Federated Learning
- 9. Lab 1. Linux Operating System
 - 9.1. Distributions
 - 9.2. Shell commands
- 10. Lab 2. NodeRED Programming Tool
 - 10.1. Programming Tool description

10.2. Examples

11. Lab 3. MQTT Publish/Subscribe Protocol

11.1. Protocol Description

11.2. Quality of Service Semantics

11.3. Example of stream Processing with Sensor Mobile, Node Red and MQTT

12. Lab 4. Stream Processing with Kafka

12.1. Definition

12.2. Installation

12.3. Kafka console commands

12.4. KAFKA Java API

12.5. Example

13. Lab 5. Blockchain with Ethereum

13.1. Definition

13.2. Example

14. Lab 6. Federated Learning

14.1. Installation

14.2. Working with Distributed FL environments.

14.3. Hands on: Example of FL with existing models.

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Lesson 1. Introduction Duration: 02:45 Lecture	Lab 1. Linux. Node Red Duration: 02:45 Laboratory assignments		
2	Lesson 2. Indirect communication. Publisher-subscriber. Lesson 3. Stream Processing Duration: 01:45 Lecture	Lab 2. MQTT Protocol Duration: 01:45 Laboratory assignments		
3	Lesson 3. Stream Processing Lesson 4. Blockchain. Ethereum. Non Fault Tolerant Approach (1) Duration: 01:45 Lecture	Lab3. MQTT Protocol Duration: 01:45 Laboratory assignments		
4	Lesson 4. Blockchain. Ethereum. Non Fault Tolerant Approach (1) Duration: 01:30 Lecture	Lab3. MQTT Protocol. Kafka Stream Processing (1h + 0,45m) Duration: 01:45 Laboratory assignments		
5	Lesson 4. Blockchain. Ethereum. Non Fault Tolerant Approach (1) Duration: 02:45 Lecture	Lab 4. Kafka Stream Processing Duration: 02:30 Laboratory assignments		EX1: (Theory and Lab) Written test Progressive assessment Presential Duration: 01:00
6	Lesson 5.Replica consistency and CAP theorem Duration: 01:30 Lecture	Lab 4. Kafka Stream Processing Duration: 01:45 Laboratory assignments		
7	Lesson 6. Raft Duration: 01:45 Lecture	Lab 5. Ethereum Blockchain Duration: 01:30 Laboratory assignments		
8	Lesson 6. Raft Lesson 7. Blockchain. Ethereum. Fault Tolerant Approach (2) Duration: 02:00 Lecture	Lab 6. Student Project MQTT/Kafka/Ethereum Duration: 01:15 Laboratory assignments		
9	Lesson 7. Blockchain. Ethereum. Fault Tolerant Approach (2) Duration: 03:30 Lecture	Student Project presentation (I) Duration: 02:00 Laboratory assignments		
10	Lesson 8. Federated Systems. Federated Learning Duration: 02:00 Lecture	Lab 7. Federated Learning Duration: 02:00 Laboratory assignments		

11	Lesson 9. Federated Systems. Duration: 00:00 Lecture	Student Project presentation (II) Duration: 02:00 Laboratory assignments		Project MQTT/Kafka/Ethereum/FL presentation Group presentation Progressive assessment Presential Duration: 03:15
12				
13				
14				
15				
16				
17				Ex1(Theory and Lab) final exam. Written test Global examination Presential Duration: 02:00 Project MQTT/Kafka/Ethereum presentation Individual presentation Global examination Presential Duration: 02:00 EX2 (Theory and Lab) Written test Progressive assessment Presential Duration: 02:00 Ex2(Theory and Lab) Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
5	EX1: (Theory and Lab)	Written test	Face-to-face	01:00	30%	4 / 10	CT.01 CE.05 CE.08 CB10
11	Project MQTT/Kafka/Ethereum/FL presentation	Group presentation	Face-to-face	03:15	30%	0 / 10	CT.01 CE.05 CE.06 CE.08 CB08 CB10 CG01 CG02
17	EX2 (Theory and Lab)	Written test	Face-to-face	02:00	40%	4 / 10	CT.01 CE.05 CE.08 CB08 CG01 CG02

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Ex1(Theory and Lab) final exam.	Written test	Face-to-face	02:00	30%	4 / 10	CT.01 CE.05 CE.06 CE.08 CB10 CG01 CG02
17	Project MQTT/Kafka/Ethereum presentation	Individual presentation	Face-to-face	02:00	30%	5 / 10	CT.01 CE.05 CE.06 CE.08 CB08 CB10 CG01 CG02

17	Ex2(Theory and Lab)	Written test	Face-to-face	02:00	40%	4 / 10	CT.01 CE.06 CE.08 CB08 CG01 CG02
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7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Ex(Theory and Lab)	Written test	Face-to-face	02:00	100%	4 / 10	CT.01 CE.05 CE.06 CE.08 CB10 CG01 CG02

7.2. Assessment criteria

EVALUACIÓN PROGRESIVA (Continuous Evaluation):

Week 4

PartialMark1= Ex1 (Theory+ Lab) * 0.3

week17

PartialMark2= Ex2 (Theory+ Lab) * 0.4

AND

IF PartialMark1

$$\text{PartialMark1} = \text{Ex1Re}(\text{Theory} + \text{Lab}) * 0.3$$

$$\text{ProjectMark} = \text{project} * 0.25 \text{ (Oral presentation + Teacher's assessment of the work done).}$$

Mark calculation:

IF **PartialMark1** >= 4 AND **PartialMark2** >= 4:

$$\text{FinalMark} = \text{PartialMark1} + \text{PartialMark2} + \text{ProjectMark}$$

The course is passed if **FinalMark** >= 5.

Explanation:

Students opting for this modality will have the chance to do the first partial (EX1) by week4. In case the mark is lower than 4, the students will have the opportunity of re-examination in the final date. The second exam (EX2) covers the rest of the syllabus (topics 4-9). Both (EX1 and Ex2) have a minimum threshold score of 4.

GLOBAL EVALUATION

week17

$$\text{PartialMark1} = \text{Ex1Re}(\text{Theory} + \text{Lab}) * 0.3$$

$$\text{PartialMark2} = \text{Ex2} (\text{Theory} + \text{Lab}) * 0.4$$

Mark calculation:

IF **PartialMark1** >= 4 AND **PartialMark2** >= 4:

$$\text{FinalMark} = \text{PartialMark1} + \text{PartialMark2} + \text{ProjectMark}$$

The course is passed if **FinalMark** >= 5.

Explanation:

Students opting for this modality will do the first partial (EX1 topics 1-3). The second exam (EX2) covers the rest of the syllabus (topics 4-9). Both (EX1 and Ex2) have a minimum threshold score of 4. The course is passed if **FinalMark** ≥ 5 .

The final mark is calculated as the weighted sum of both partial exams (if both are ≥ 4) and the weighted score from the project.

EXTRA-ORDINARY EVALUATION

FinalMark = Ex (Theory + Lab)

The course is passed if **FinalMark** ≥ 5 .

Explanation:

At this stage, the evaluation consists of an exam of the whole syllabus which will be 100% of the course grade.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Distributed Systems, concepts and design, 4th Edition. G. Coulouris. J. Dollimore, T. Kindberg, G. Blair. Addison Wesley, 2012.	Bibliography	

Introduction to Reliable and Secure Distributed Programming Authors: Cachin, Christian, Guerraoui, Rachid, Rodrigues, Luís. Springer (2011)	Bibliography	
Communication and Agreement Abstractions for Fault Tolerant Asynchronous Distributed Systems. Michel Raynal. Morgan & Claypool Publishers 2010	Bibliography	
Making Sense of Stream Processing. By Martin Kleppmann Publisher: O'Reilly Released: May 2016	Bibliography	
Raft Paper	Bibliography	D. Ongaro, J. Ousterhout. In Search of an Understandable Consensus Algorithm. USENIX Annual Technical Conference (ATC), Philadelphia, PA, 2014
Apache Kafka Documentation	Web resource	https://kafka.apache.org/
Moodle de la asignatura	Web resource	moodle upm
Laboratorio de ordenadores con sistema operativo tipo Unix.	Equipment	
MQTT	Web resource	http://mqtt.org/ MQTT is a machine-to-machine (M2M)/"Internet of Things" connectivity protocol. It was designed as an extremely lightweight publish/subscribe messaging transport. It
Zookeeper	Web resource	https://zookeeper.apache.org/
Mastering Ethereum. Andreas Publisher: Antonopoulos O'Reilly. and Gavin 2018. Wood.	Bibliography	
Blockchain Paper	Bibliography	From blockchain consensus back to Byzantine consensus. Vincent Gramoli. Future Generation Computer Systems, V107, June 2020.

9. Other information

9.1. Other information about the subject

UPM Teams Tool will be used as the default communication tool with the student. Please, check

<https://www.upm.es/UPM/ServiciosTecnologicos/Office365>.